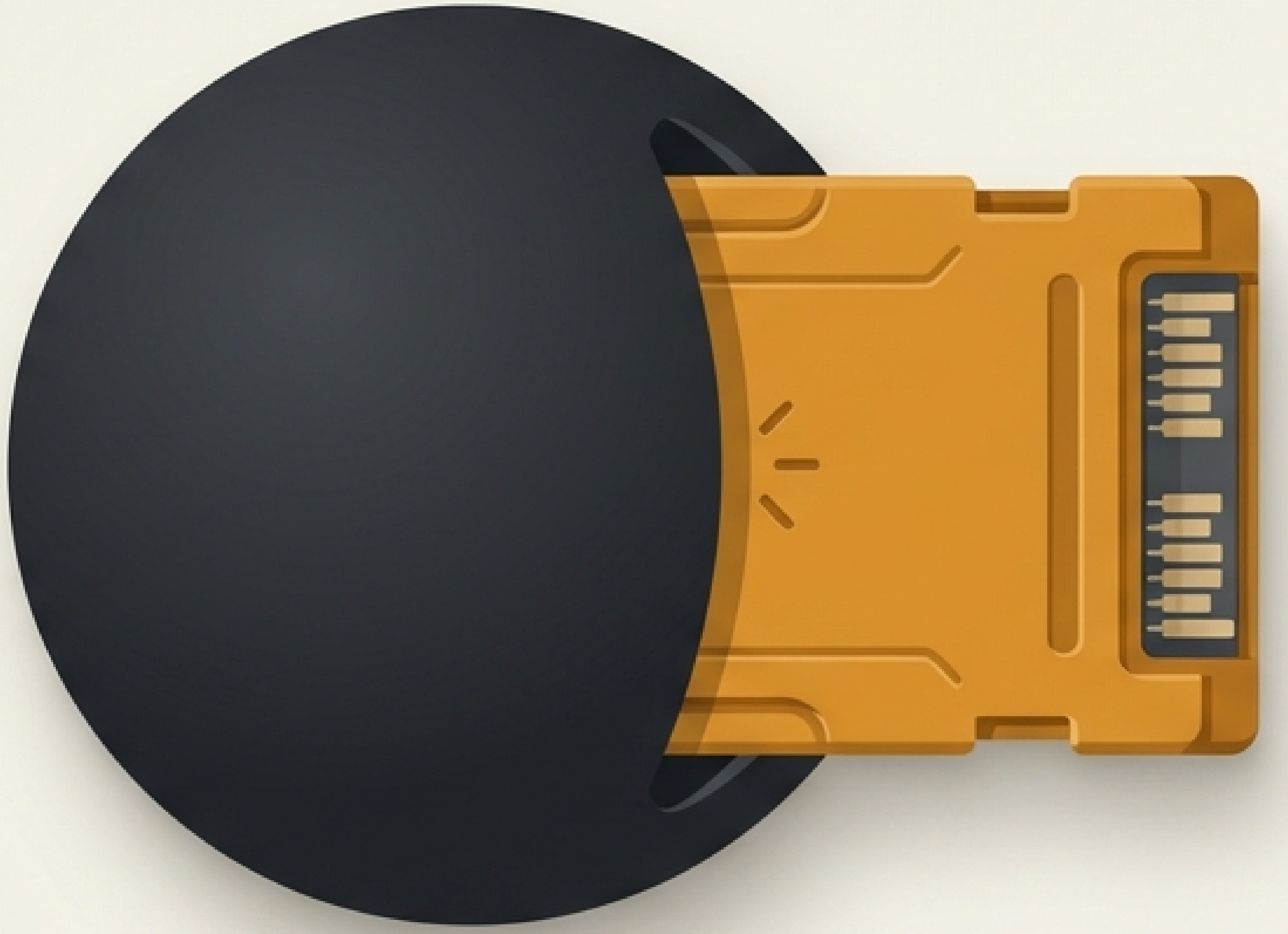
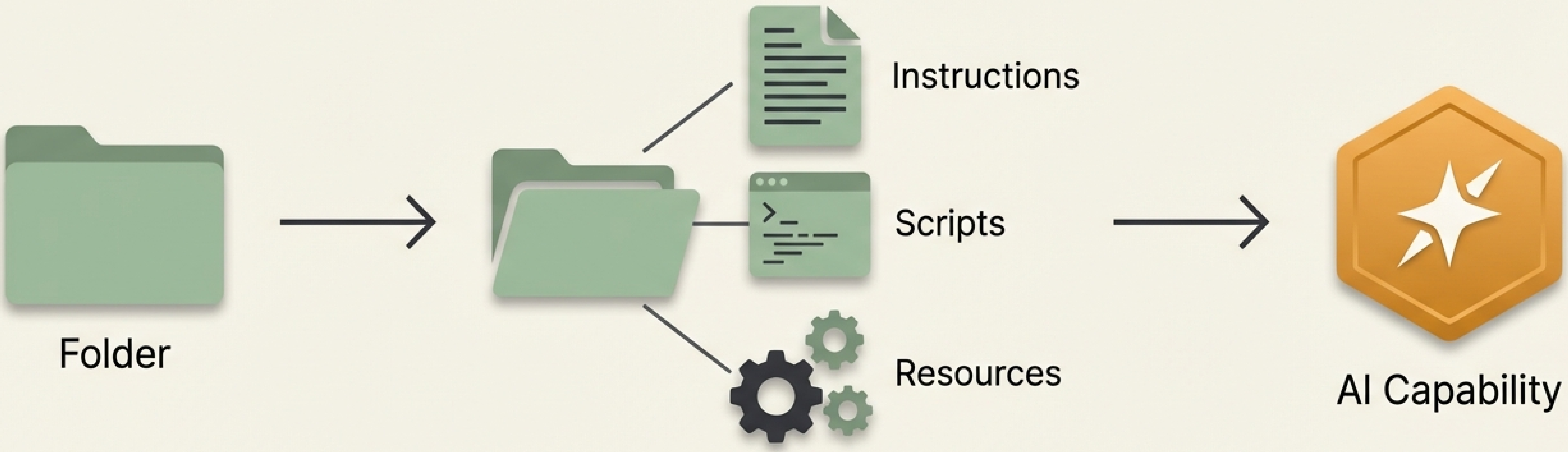
Supercharging AI with Agent Skills

Give AI agents new capabilities, domain expertise, and procedural knowledge on demand. Write once, use everywhere.



What is an Agent Skill?

- An Agent Skill is a self-contained, version-controlled package of procedural knowledge.
- It bundles instructions, scripts, and resources together.
- Allows an AI to reliably follow a specific workflow, bridging the gap between native LLM knowledge and real-world execution.



The Challenge of Context

Project Instructions

Always-on, general information applied to every single prompt in a project.

Prompt Files

Short, reusable text snippets for common, repetitive queries.

Custom Agents

Specific, locked-in workflows that dictate the agent's overall behavior.

Agent Skills

Modular, on-demand capabilities. Bundles multiple files (markdown, scripts, templates) into a single, triggerable tool.

The Directory Structure

```
my-skill/
├── SKILL.md
├── scripts/
│   └── extract.py
├── references/
│   └── schema.json
└── assets/
    └── template.md
```

- `SKILL.md` (Required): The core metadata (Name, Description) and the actual instructions for the AI to follow.
- `scripts/` (Optional): Executable code (Python, Node, Bash) to perform deterministic actions.
- `references/` (Optional): Documentation, schemas, or specific context files.
- `assets/` (Optional): Templates and resources to format the AI's output.

The Engine of Progressive Disclosure

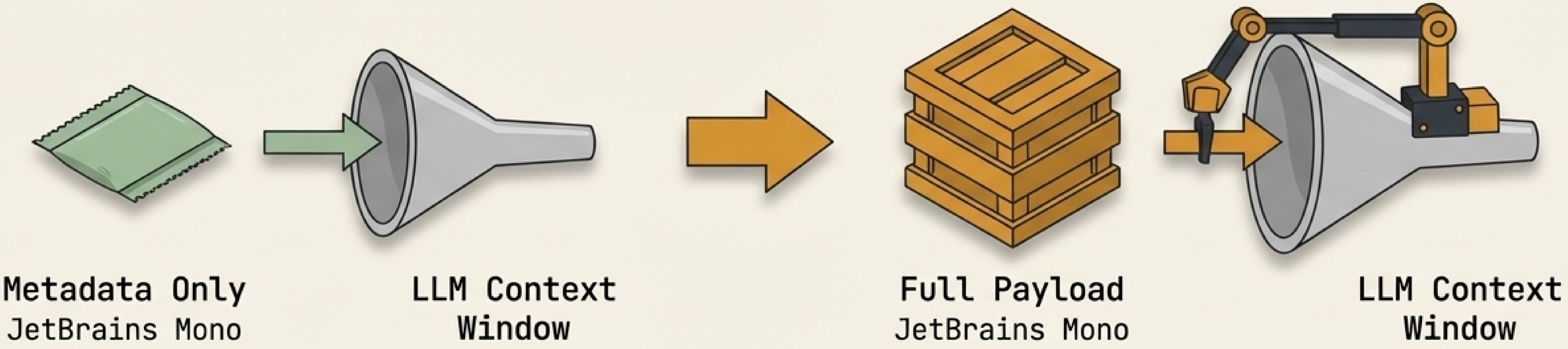
Skills manage context efficiently by loading in stages.

Stage 1 (Discovery)

The AI only sees the skill's name, description, and file path in its system prompt.

Stage 2 (Execution)

When the AI decides the skill is needed, it explicitly reads the full SKILL.md instructions and scripts into the context window.



Implicit vs. Explicit Invocation

Explicit Invocation

The user commands the AI directly.

```
> /skills
> User: Please run $pdf-extractor
```

Implicit Invocation

The AI reads the user's intent and automatically matches it to a skill's description.

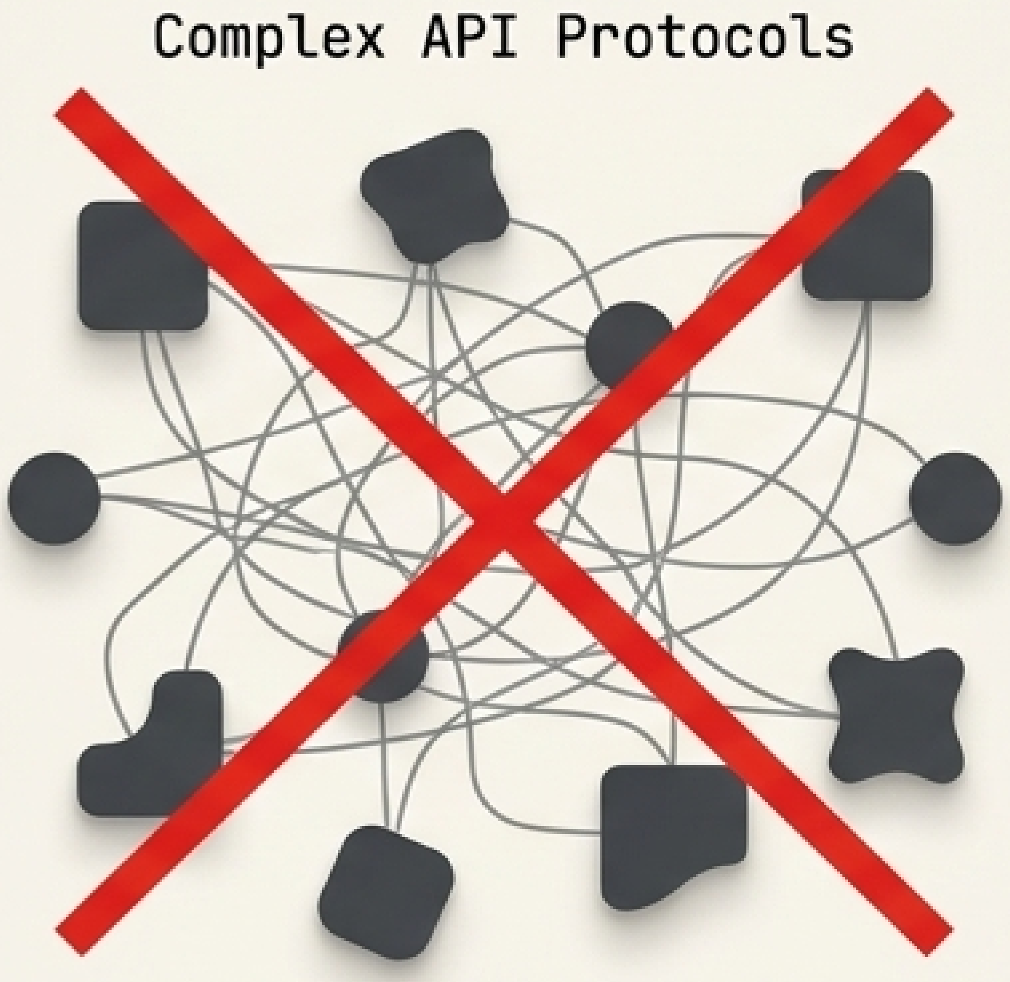
User: Can you find the invoice total in this document?

Automatically matched to PDF Skill

Pro Tip: Because implicit matching relies entirely on the description field in SKILL.md, boundaries and triggers must be explicitly defined.

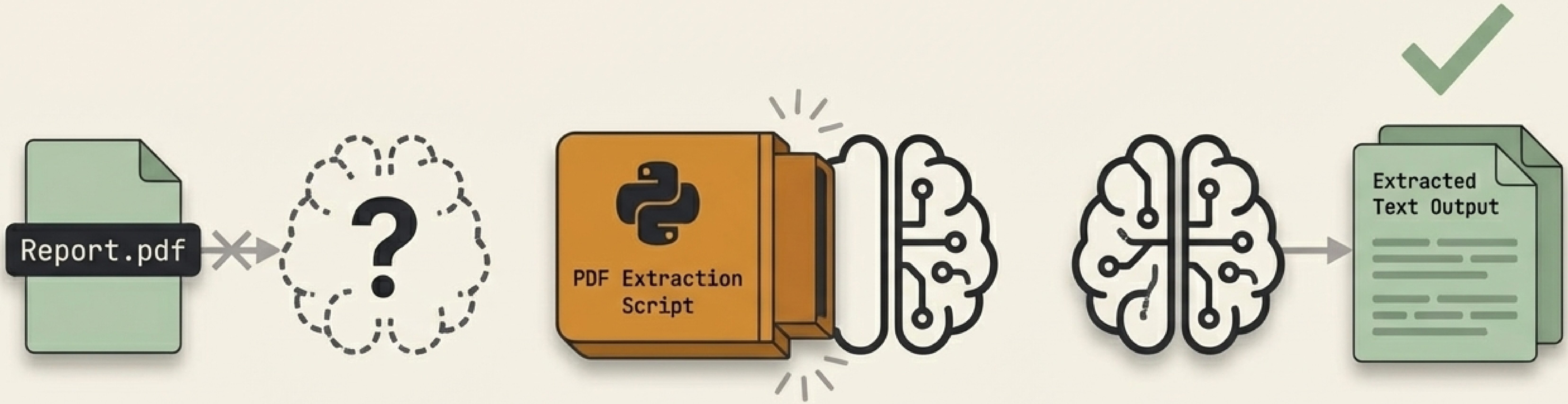
The UNIX Philosophy for AI

- Complex protocols often create friction.
- CLI applications are highly structured, text-based, and inherently LLM-friendly.
- Programs should be written to handle text streams, because that is a universal interface.



Real-World Power: Reading PDFs

The Problem: Native AI agents often struggle to parse or read local PDF files.
The Solution: Equipping a PDF skill that bundles Python extraction scripts inside the folder.
The Result: The AI learns to bridge its own native capability gap, automatically running the script to extract text.

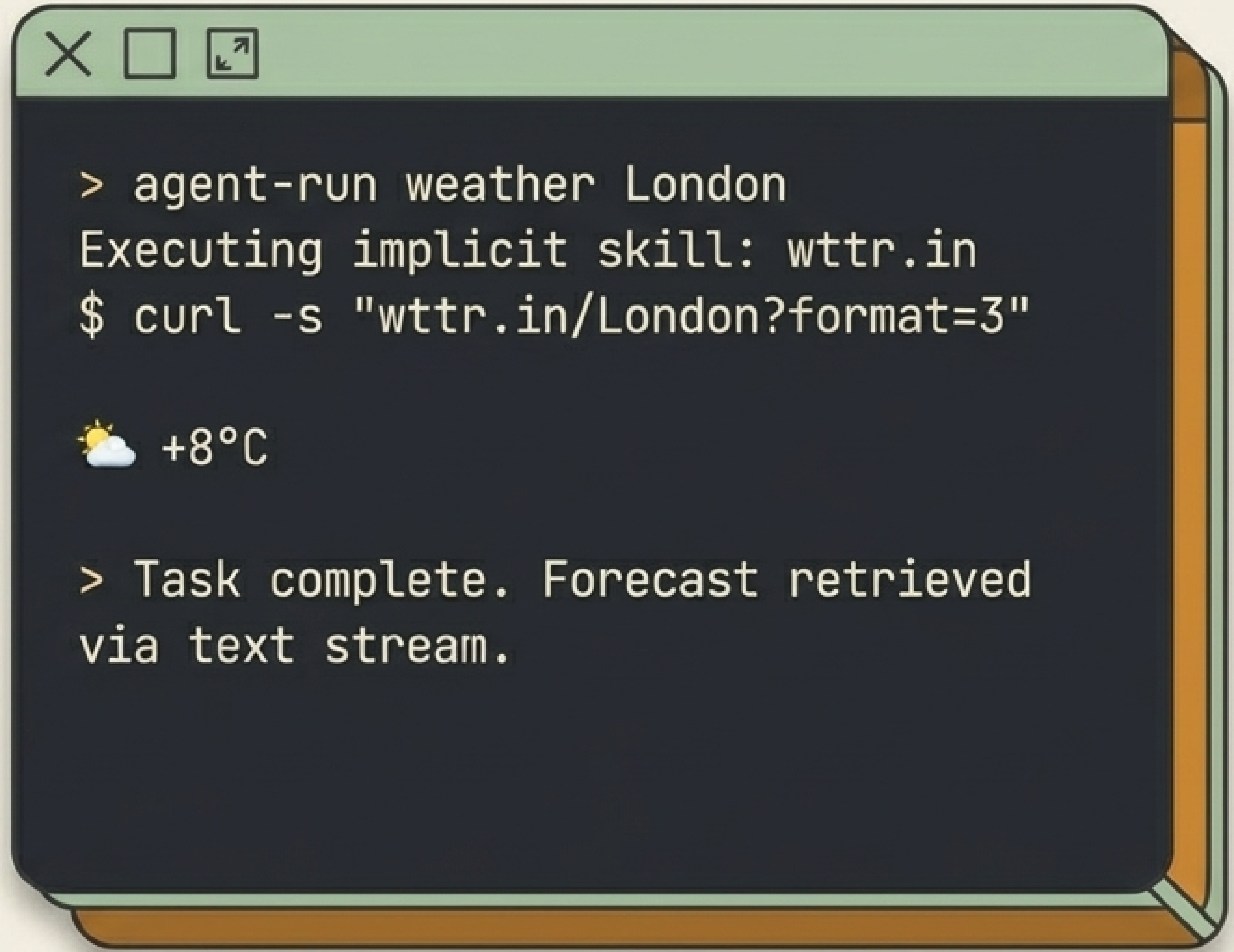


Elegant Simplicity: Instant Weather

Skills don't need to be massive applications; they can wrap elegant CLI commands.

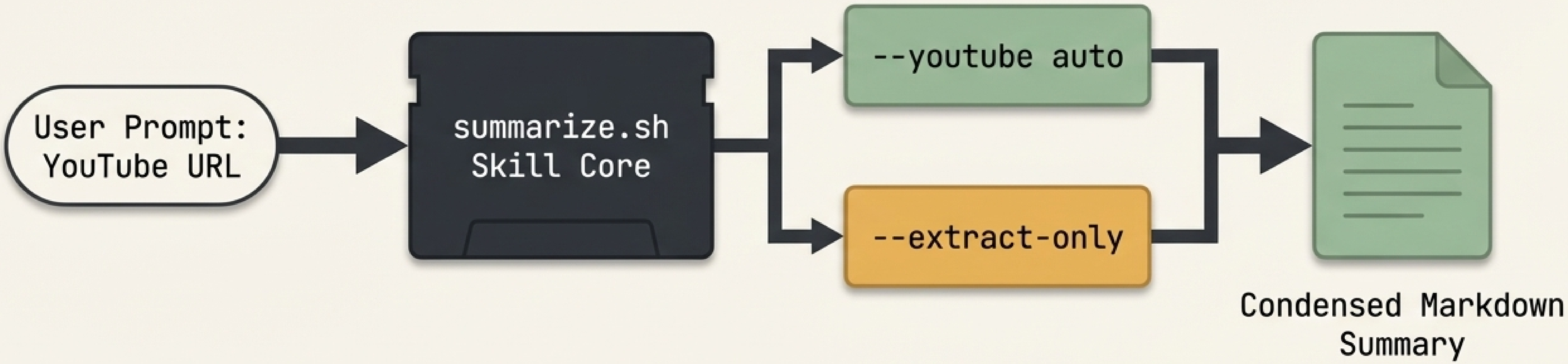
Allows the AI to fetch local forecasts without API keys.

Proves that text streams are the ultimate interface for LLMs.



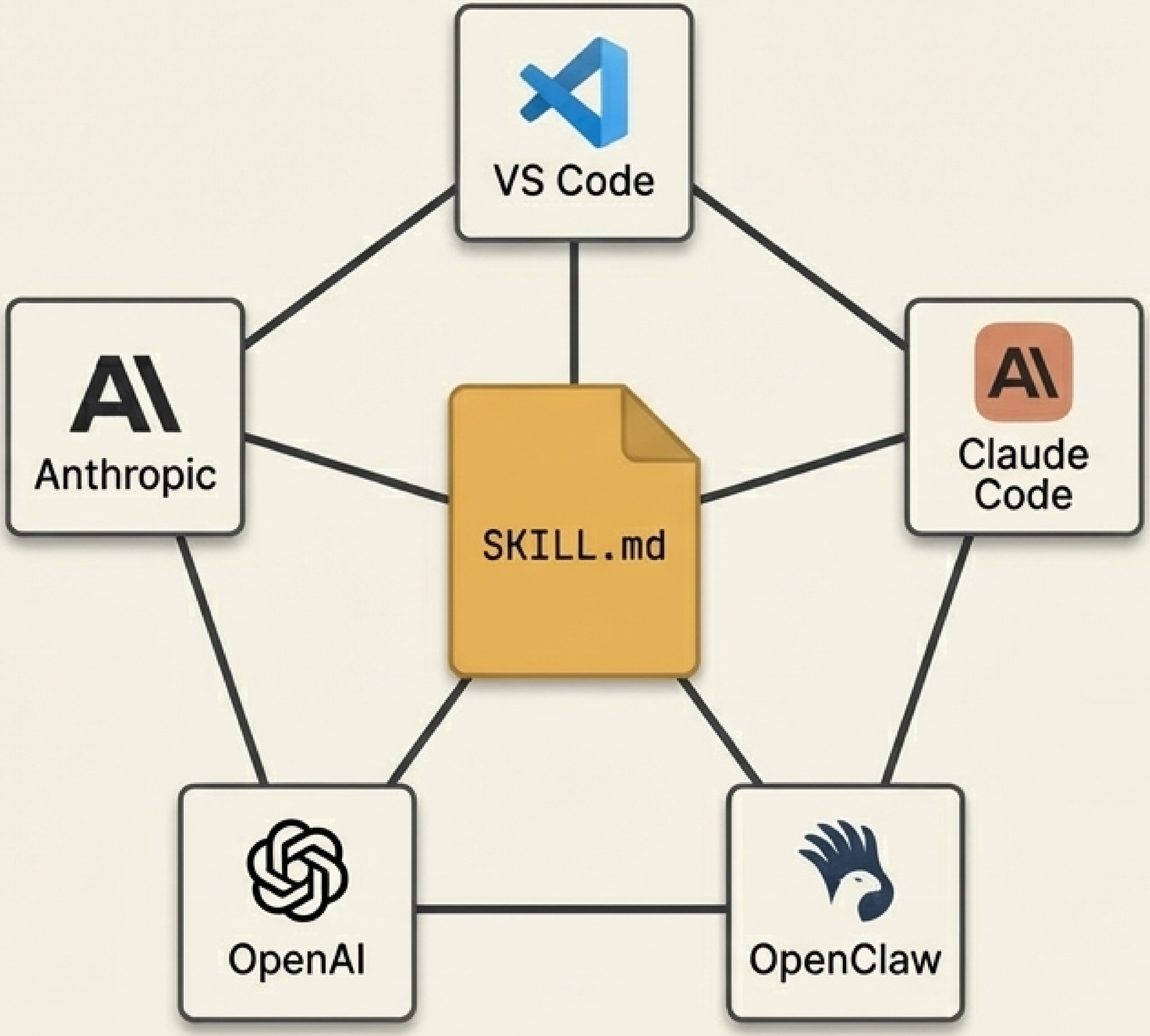
Complex Workflows: The Summarize Tool

- The **summarize.sh** skill acts as a fast CLI to extract text from URLs, local files, and YouTube links.
- **Dynamic Flags:** The AI learns to use flags like `--youtube auto` or `--extract-only` based entirely on the user's intent.
- **Built-in Fallbacks:** Gracefully handles missing tools by utilizing fallbacks like Apify for YouTube.



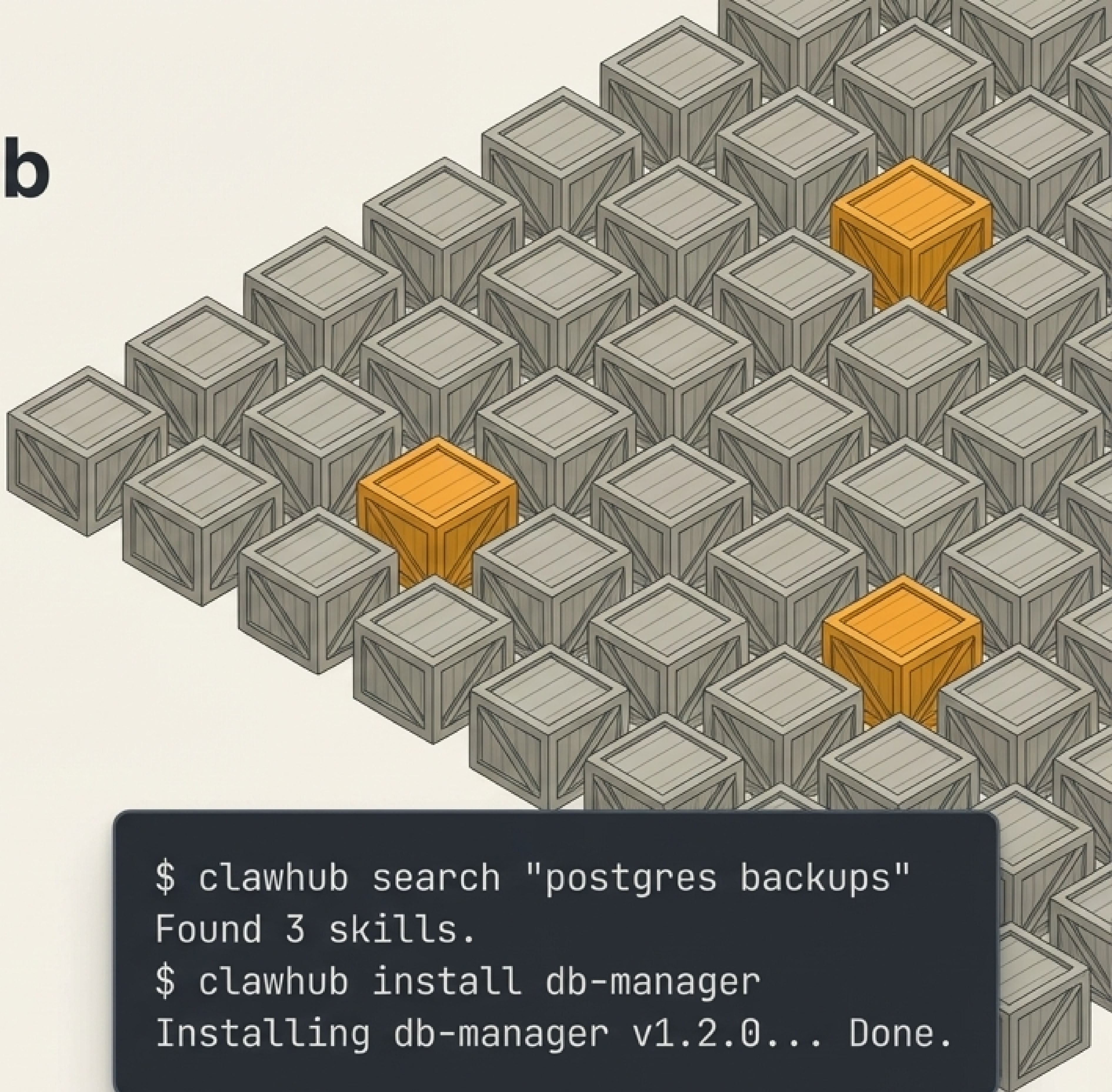
An Open Standard Ecosystem

- Agent Skills are not locked to a single platform or vendor.
- Originally developed by Anthropic and released as an open standard.
- Adopted by a growing number of AI products and agents.
- Allows developers to capture organizational knowledge in portable packages and reuse them anywhere.



Scaling Up with ClawHub

- ClawHub is the public registry for OpenClaw skills—like npm for AI capabilities.
- Discovery: Search for skills via plain language and vector embeddings.
- Management: Install, update, and publish versioned bundles via the CLI.
- Integrity: Tracks installations via .clawhub/lock.json and uses content hashes safely.



```
$ clawhub search "postgres backups"
Found 3 skills.
$ clawhub install db-manager
Installing db-manager v1.2.0... Done.
```

The Future of Modular AI

- Agent Skills replace complex protocols with simple, version-controlled text and scripts.
- They solve context window bloat through progressive disclosure.
- They empower developers to teach any AI, any workflow, at any time.